

Chapter 5 Assignment Due date: June 01, 2026 (Mon)

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5.1

Sol.

The EWMA is defined by

$$E_n = \lambda X_n + (1 - \lambda)E_{n-1}, \quad 0 < \lambda \leq 1$$

Expanding E_n , we obtain

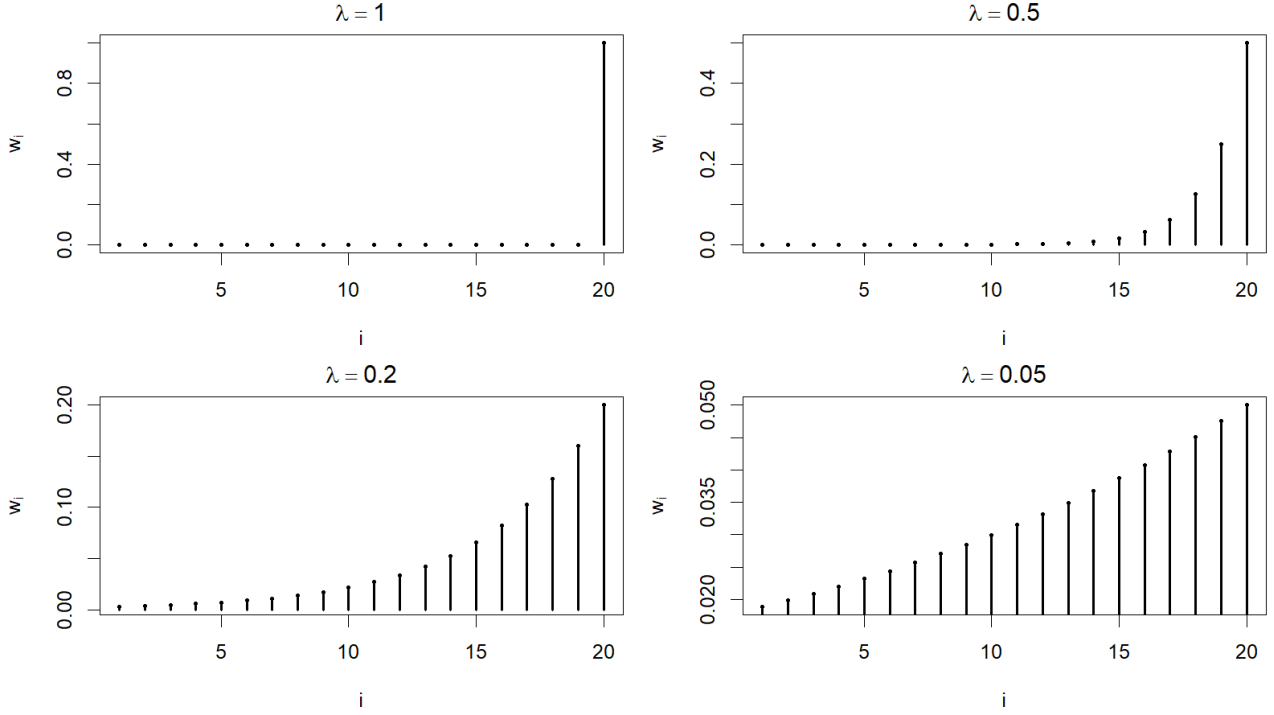
$$E_n = \sum_{i=1}^n \lambda(1 - \lambda)^{n-i} X_i + (1 - \lambda)^n \mu_0$$

Since $n = 20$ and the weight of X_i in E_{20} is

$$w_i = \lambda(1 - \lambda)^{20-i}, \quad i = 1, 2, \dots, 20$$

Then we compute weights for $\lambda = 1, 0.5, 0.2, 0.05$ are:

i	Observation	$\lambda = 1$	$\lambda = 0.5$	$\lambda = 0.2$	$\lambda = 0.05$
1	X_1	0	0.000001	0.002882	0.018868
2	X_2	0	0.000002	0.003603	0.019861
3	X_3	0	0.000004	0.004504	0.020906
4	X_4	0	0.000008	0.005630	0.022006
5	X_5	0	0.000015	0.007037	0.023165
6	X_6	0	0.000031	0.008796	0.024384
7	X_7	0	0.000061	0.010995	0.025667
8	X_8	0	0.000122	0.013744	0.027018
9	X_9	0	0.000244	0.017180	0.028440
10	X_{10}	0	0.000488	0.021475	0.029937
11	X_{11}	0	0.000977	0.026844	0.031512
12	X_{12}	0	0.001953	0.033554	0.033171
13	X_{13}	0	0.003906	0.041943	0.034917
14	X_{14}	0	0.007813	0.052429	0.036755
15	X_{15}	0	0.015625	0.065536	0.038689
16	X_{16}	0	0.031250	0.081920	0.040725
17	X_{17}	0	0.062500	0.102400	0.042869
18	X_{18}	0	0.125000	0.128000	0.045125
19	X_{19}	0	0.250000	0.160000	0.047500
20	X_{20}	1	0.500000	0.200000	0.050000



From the figures, we can know that, when $\lambda = 1$, the EWMA statistic becomes

$$E_n = X_n$$

Therefore, for $n = 20$, only X_{20} receives weight 1, and all previous observations receive weight 0. For $0 < \lambda < 1$, the weight of X_i in E_{20} is

$$w_i = \lambda(1 - \lambda)^{20-i}$$

Since $0 < 1 - \lambda < 1$, the term $(1 - \lambda)^{20-i}$ is smaller for older observations and larger for newer observations. Hence, the weights increase from X_1 to X_{20} .

5.4

Sol.

Consider the EWMA statistic defined in (5.1),

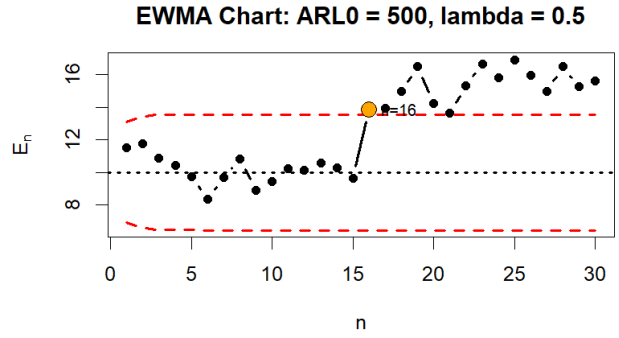
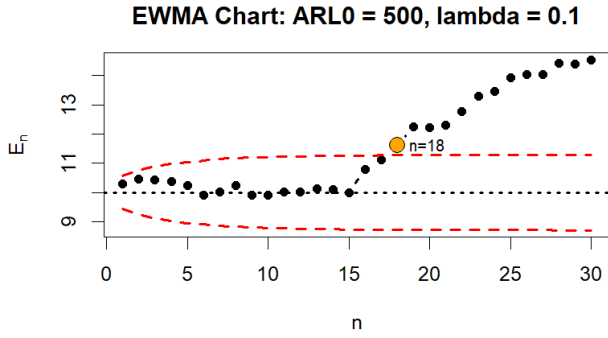
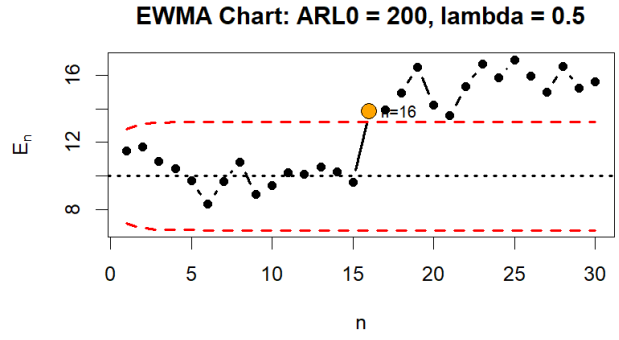
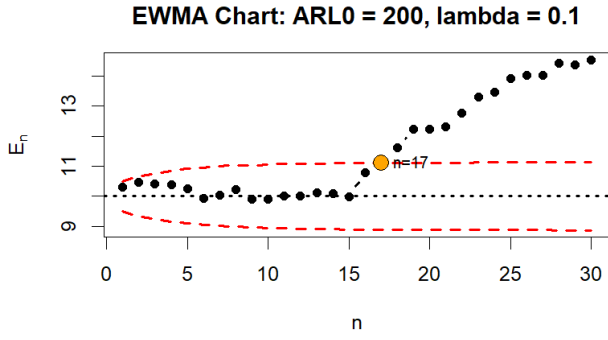
$$E_n = \lambda X_n + (1 - \lambda)E_{n-1},$$

and the two-sided EWMA control limits defined in (5.7)

$$U_n = \mu_0 + \rho \sqrt{\frac{\lambda}{2 - \lambda} [1 - (1 - \lambda)^{2n}]} \sigma, \quad L_n = \mu_0 - \rho \sqrt{\frac{\lambda}{2 - \lambda} [1 - (1 - \lambda)^{2n}]} \sigma$$

We construct two-sided EWMA charts. The first signal times are summarized as follows:

Case	ARL_0	λ	ρ	First signal
(i)	200	0.1	2.454	17
(ii)	200	0.5	2.777	16
(iii)	500	0.1	2.814	18
(iv)	500	0.5	3.071	16



For the same ARL_0 , a larger λ gives more weight to the current observation and makes the control limits wider. , but it also makes the EWMA statistic more sensitive to new observations. Therefore, the chart with $\lambda = 0.5$ signals earlier than the chart with $\lambda = 0.1$.

For the same λ , a larger ARL_0 requires a larger value of ρ , which makes the control limits wider. Therefore, the chart tends to signal later.

However, when $\lambda = 0.5$, both $ARL_0 = 200$ and $ARL_0 = 500$ signal at $n = 16$. This happens because the upward shift is large enough that even the wider control limits for $ARL_0 = 500$ are crossed at the same time.

5.6

Sol.

We compute ARL_0 , and the results are summarized as follows:

Case	λ	ρ	ARL_0
(i)	0.1	1	10.422
(ii)	0.1	2	73.276
(iii)	0.5	1	3.786
(iv)	0.5	2	26.452

From the results, for the same λ , a larger ρ gives a larger ARL_0 . This is because ρ controls the width of the control limits. When ρ increases, the control limits become wider, so false alarms occur less often.

For the same ρ , a larger λ gives a smaller ARL_0 . This is mainly because λ controls the smoothing effect of the EWMA statistic. When λ is larger, the chart gives more weight to the current observation and behaves more like a Shewhart chart. Therefore, under the IC condition, the statistic fluctuates more and is more likely to cross the control limits, leading to a smaller ARL_0 .

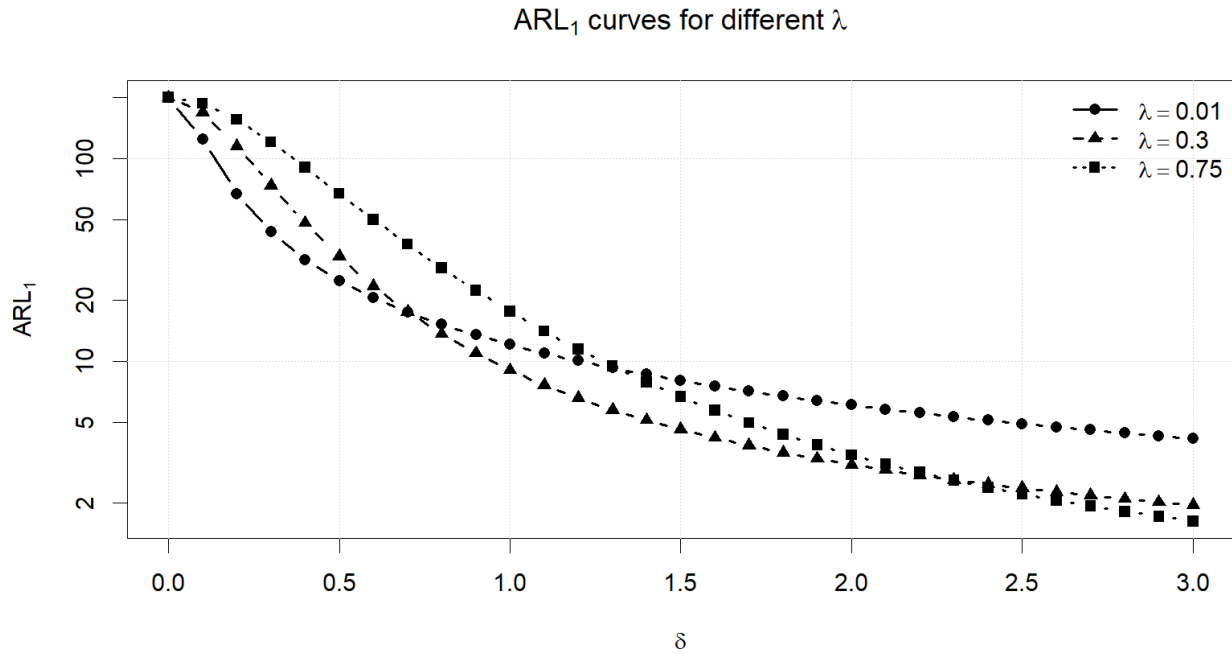
5.8

Sol.

The computed ρ values are

λ	ρ
0.01	1.500
0.30	2.713
0.75	2.802

where each ρ is chosen so that $ARL_0 = 200$.



From the plot, all three curves start from $ARL_0 = 200$ when $\delta = 0$. As the shift size δ increases, ARL_1 decreases for all three values of λ . This is reasonable because a larger mean shift is easier to detect.

For small shifts, such as $\delta = 0.1$, the chart with $\lambda = 0.01$ has the smallest ARL_1 . This indicates that a smaller λ is more effective for detecting small persistent shifts, because it gives stronger smoothing and has a longer memory.

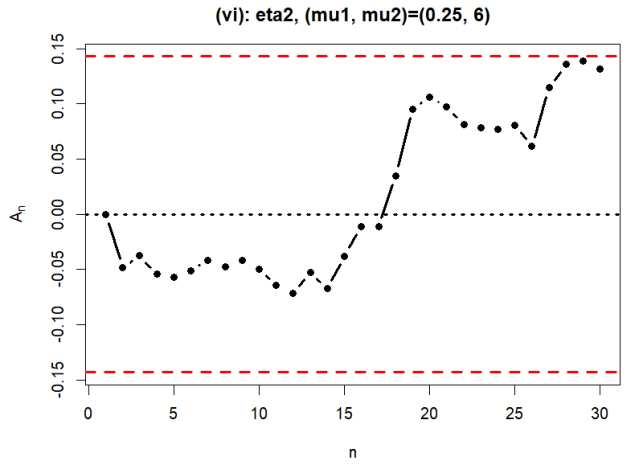
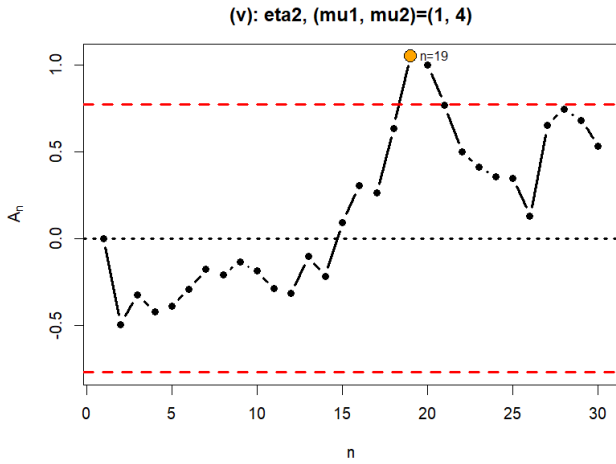
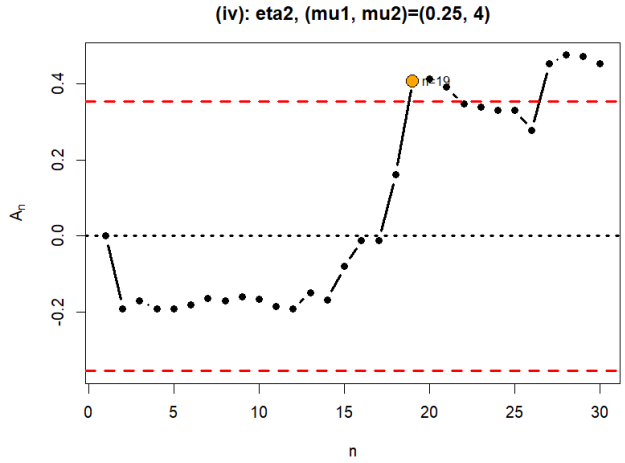
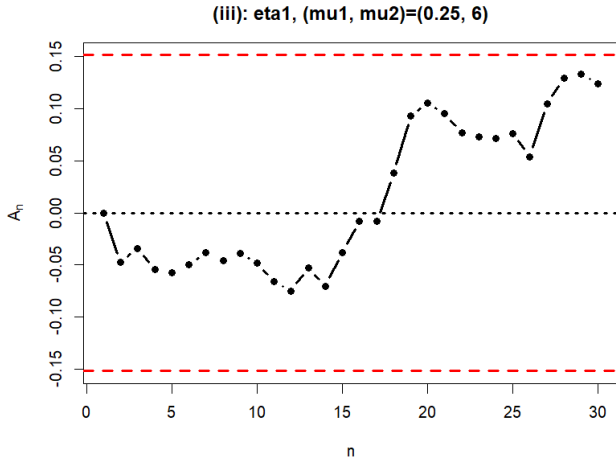
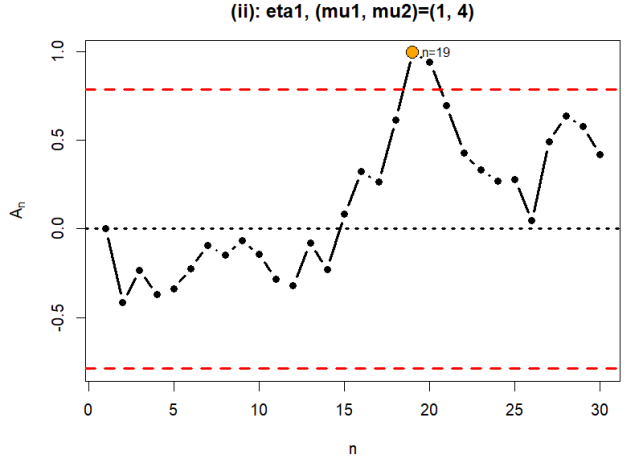
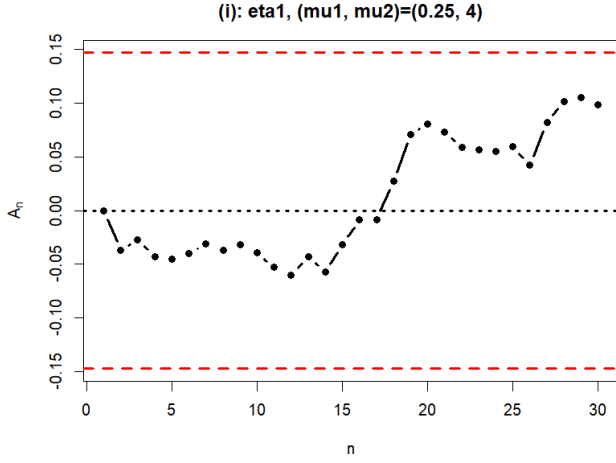
For large shifts, such as $\delta = 3.0$, the chart with $\lambda = 0.75$ has the smallest ARL_1 . This indicates that a larger λ is more effective for detecting large shifts, because it gives more weight to the current observation and reacts more quickly.

Therefore, the choice of λ should depend on the size of the mean shift that we want to detect. A smaller λ is more suitable for detecting small and persistent shifts, because it gives more smoothing and uses more historical information. A larger λ is more suitable for detecting large shifts, because it gives more weight to new observations and reacts more quickly.

5.20

Sol.

Case	η	μ_1	μ_2	h	λ	u	First signal
(i)	η_1	0.25	4	0.1471	0.0162	2.7459	NA
(ii)	η_1	1.00	4	0.7874	0.1813	2.5752	19
(iii)	η_1	0.25	6	0.1515	0.0207	4.2537	NA
(iv)	η_2	0.25	4	0.3542	0.0188	12.6145	19
(v)	η_2	1.00	4	0.7697	0.1359	10.6114	19
(vi)	η_2	0.25	6	0.1430	0.0168	48.7509	NA



From the results, cases (ii), (iv), and (v) give signals at $n = 19$, while cases (i), (iii), and (vi) do not give any signal within the first 30 observations.

This indicates that the detection result depends on the choice of the adaptive score function $\eta(e_n)$ and the design parameters (μ_1, μ_2) . In this dataset, the observations around $n = 18$ and $n = 19$, especially $X_{18} = 2.2$ and $X_{19} = 2.7$, create a relatively large upward movement. Therefore, the more sensitive settings can detect this upward shift at $n = 19$.

For η_1 , case (ii) gives a signal, but cases (i) and (iii) do not. This is mainly because case (ii) has a larger λ , so the adaptive EWMA statistic reacts more quickly to recent observations.

For η_2 , cases (iv) and (v) give signals, while case (vi) does not. This shows that different choices of (μ_1, μ_2) and u can change the sensitivity of the chart. In particular, case (vi) has a very large u , so the statistic changes more smoothly and does not cross the control limit in these 30 observations.

However, whether the chart can detect the shift depends on the choice of $\eta(e_n)$ and the parameter settings.